

Project 4 - Proposal

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Topic

This is a research project focused on image classification of recyclable materials. Two different machine learning models will be trained and evaluated on the same dataset, and their performances will be compared to determine which approach is more effective for this classification task.

Problem Statement

Proper recycling remains a widespread challenge due to difficulties in accurately identifying recyclable materials. This project addresses that problem by training machine learning models to automatically classify types of recyclable materials from 150x150 color images using the UNF AI Lab recyclable material dataset. Two models will be implemented and compared: a custom Convolutional Neural Network built from scratch, and a pretrained ResNet model fine-tuned using transfer learning. The goal is to evaluate which approach yields better classification accuracy, and to analyze the trade-offs between building a model from scratch versus leveraging a pretrained one.

Planned Activities and Timeline

April 10 - Submit proposal - Matt

April 11-12 - Download dataset, set up Google Colab environment, explore images - Both

April 13-14 - Build and train CNN model - Matt

April 15-16 - Implement the ResNet transfer learning model - Christian

April 17 - Submit email progress report - Christian

April 18-19 - Compare model results - Both

April 20-22 - Write final report - Matt

April 23-24 - Finalize up source code, write README - Christian

April 26 - Package zip file and submit final project to canvas - Matt